Tutorial 3 Solution

Problem 1

(a) Let

$$f : \mathbb{R}^3_{++} \to \mathbb{R}, (x_1, x_2, x_3) \mapsto \frac{3x_1^2 + e^{x_2}}{2x_3}.$$

Compute all partial derivatives.

(b) Evaluate the partial derivative of f with respect to x_3 at the point $(1, 2, 3)$.

Solution:

(a) The partial derivatives are

$$\frac{\partial f(x_1, x_2, x_3)}{\partial x_1} = \frac{3x_1}{x_3}$$
$$\frac{\partial f(x_1, x_2, x_3)}{\partial x_2} = \frac{e^{x_2}}{2x_3}$$
$$\frac{\partial f(x_1, x_2, x_3)}{\partial x_3} = \frac{-3x_1^2 - e^{x_2}}{2x_3^2}$$

To derive them, simply hold all other variables constant. This allows you to treat the function as a function of one variable—the one you are differentiating with respect to—and apply the rules of differentiation for functions of one variables at the beginning of Lecture 3 slides.

(b) We obtain

$$\frac{\partial f(1, 2, 3)}{\partial x_3} = \frac{-3 \cdot 1^2 - e^2}{2 \cdot 3^2} = \frac{-3 - e^2}{18}.$$

Problem 2

You are given the function $f : D \to \mathbb{R}, (x, y) \mapsto e^{\frac{x}{y}} + \ln(x^2 + 1)$, where D is the largest subset of $\mathbb{R}^2$ on which f is well-defined.

(a) Determine D.

Solution: The domain of f is $D = \{(x, y) \in \mathbb{R}^2 : \ y \neq 0\}$: if $y = 0$, $e^{\frac{x}{y}}$ is not well-defined.

(b) Recall that the gradient $\nabla f(x,y)$ of f at (x,y) is defined as $\nabla f(x,y) = \left(\frac{\partial f(x,y)}{\partial x}, \frac{\partial f(x,y)}{\partial y}\right)$. Calculate the gradient of f at every point (x,y).

Solution: We first determine f's partial derivatives as

$$\frac{\partial f(x,y)}{\partial x} = \frac{1}{y}e^{\frac{x}{y}} + \frac{2x}{x^2+1},$$
$$\frac{\partial f(x,y)}{\partial y} = -\frac{x}{y^2}e^{\frac{x}{y}}.$$

Then, the gradient of f at (x,y) is

$$\nabla f(x,y) = \left(\frac{1}{y}e^{\frac{x}{y}} + \frac{2x}{x^2+1}, -\frac{x}{y^2}e^{\frac{x}{y}}\right)$$

(c) Show that all points of the form $(0,y)$, $y \neq 0$, are located on the same level curve.

Solution: For every point $(0,y)$ such that $y \neq 0$

$$f(0,y) = e^{\frac{0}{y}} + \ln(0+1) = e^0 + \ln(1) = 1 + 0 = 1.$$

Hence, $f(0,y)$ does not depend on y, and all points of the form $(0,y)$ are on the level curve of level 1.

(d) (**Harder**) Determine, *without checking the definition directly*, whether $f(x,y)$ is continuously differentiable at the point $(1,1)$.

Solution: Recall a theorem in the lecture saying that if a function is continuously partially differentiable, then it is (totally) differentiable. It now suffices to check that both $\frac{\partial f(x,y)}{\partial x}$ and $\frac{\partial f(x,y)}{\partial y}$ are continuous at the point $(1,1)$. To show continuity, we use the properties of continuous function described in the slides of Lecture 2.

Observe that both $\frac{\partial f(x,y)}{\partial x}$ and $\frac{\partial f(x,y)}{\partial y}$ are well-defined around $(1,1)$, and

$$\frac{\partial f(x,y)}{\partial x} = \frac{1}{pr_2} \cdot e^{\frac{pr_1}{pr_2}} + \frac{2pr_1}{(pr_1)^2+1};$$

because e^x, pr_1, and pr_2 are continuous, and the relevant operations (sum, product, quotient, and composition) preserve continuity, we conclude that $\frac{\partial f(x,y)}{\partial x}$ is continuous at $(1,1)$. Similarly,

$$\frac{\partial f(x,y)}{\partial y} = -\frac{pr_1}{(pr_2)^2} \cdot e^{\frac{pr_1}{pr_2}},$$

which is also continuous by an analogous argument.

Problem 3

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto \frac{xy}{x^2+y^2+1}$. Determine $\frac{\partial f(1,1)}{\partial x}$ directly from the definition of the partial derivative, that is, by calculating the appropriate limit.

Solution: We obtain

$$\begin{aligned}
\frac{\partial f(1,1)}{\partial x} &= \lim_{n\to\infty} \frac{f(1+h_n, 1) - f(1,1)}{h_n} \\
&= \lim_{n\to\infty} \frac{\frac{1+h_n}{2+(1+h_n)^2} - \frac{1}{3}}{h_n} \\
&= \lim_{n\to\infty} \frac{3 + 3h_n - 2 - (1+h_n)^2}{3h_n(2+(1+h_n)^2)} \\
&= \lim_{n\to\infty} \frac{h_n(1-h_n)}{3h_n(2+(1+h_n)^2)} \\
&= \lim_{n\to\infty} \frac{(1-h_n)}{3(2+(1+h_n)^2)} = \frac{1}{9}.
\end{aligned}$$

You are encouraged to check that if we derive the partial derivatives using the rules of one-variable calculus by "holding other variables constant," we get the same result.

Problem 4

The production function of a firm is $f : \mathbb{R}^2_+ \to \mathbb{R}, (K, L) \mapsto 4K^{\frac{3}{4}}L^{\frac{1}{4}}$. Currently, the firm is using 10,000 units of capital (K) and employing 625 workers (L). The CEO of the firm would like to know how the production of output $(f(L, K))$ changes when the capital is increased by one unit and one more worker is employed. Can you help her?

Solution: In order to compute the change in the output production level, we can use derivatives (keep in mind that this approach provides an approximation of the exact answer). If capital changes by $dK = 1$ and labor by $dL = 1$, then the output will change by

$$\Delta f \approx df = \frac{\partial f(10000, 625)}{\partial K} \cdot dK + \frac{\partial f(10000, 625)}{\partial L} \cdot dL.$$

From

$$\begin{aligned}
\frac{\partial f(K,L)}{\partial K} &= 4\frac{3}{4}K^{-1/4}L^{1/4} = \frac{3}{4}\frac{4K^{3/4}L^{1/4}}{K} = \frac{3}{4}\frac{f(K,L)}{K}, \\
\frac{\partial f(K,L)}{\partial L} &= 4\frac{1}{4}K^{3/4}L^{-3/4} = \frac{1}{4}\frac{4K^{3/4}L^{1/4}}{L} = \frac{1}{4}\frac{f(K,L)}{L},
\end{aligned}$$

and

$$f(10000, 625) = 4(10000)^{3/4}(625)^{1/4} = 4 \cdot 1000 \cdot 5 = 20000$$

we obtain

$$\frac{\partial f(10000, 625)}{\partial K} = \frac{3}{4}\frac{20000}{10000} = \frac{3}{2},$$
$$\frac{\partial f(10000, 625)}{\partial L} = \frac{1}{4}\frac{20000}{625} = 8,$$

and can conclude that the production of output increases by

$$\Delta f \approx df = \frac{3}{2} \cdot dK + 8 \cdot dL = 9.5$$

units.

Alternatively, we can compute the value of the production function for the old and the new values of K and L and compute the difference. Currently, the firm employs $K = 10000$ units of capital, $L = 625$ units of labor, and, as we saw above, produces $f(10000, 625) = 20000$ units of output. With the increased amounts of capital and labor, the firm would produce

$$f(10001, 626) = 4(10001)^{3/4}(626)^{1/4} \approx 20009.4958$$

units of output. Hence, increasing the input of both capital and labor by 1 unit each increases output by approximately 9.4958 units.

(Just looking at this particular question, it may seem that the second approach is actually easier, and that our efforts to understand derivatives are not paying off. However, soon, when we discuss how to optimize [= maximize or minimize] functions, this changes dramatically. For optimization problems, approaches using derivatives will be "easy," while those relying on just computing different values for f will be cumbersome or even not viable.)

Problem 5 (Extra Practice Problem)

For each of the following functions f, calculate the partial derivatives, and evaluate them at the point $(1, 1)$.

(a) $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto \ln(x^2 + y^2 + 1)$

(b) $f : \mathbb{R}^2_{++} \to \mathbb{R}, (x, y) \mapsto \sqrt{xy + y^3}$

Solution:

(a) To find $\frac{\partial f(x,y)}{\partial x}$, we treat y as a constant, and use the chain rule:

1. Take the derivative of the outer function, $\ln(u)$, where $u \equiv x^2 + y^2 + 1$, which is $\frac{1}{u}$.

2. Multiply by the derivative of the inner function with respect to x, which is $\frac{\partial(x^2+y^2+1)}{\partial x} = 2x$.

Putting it together, the partial derivative with respect to x is

$$\frac{\partial f(x,y)}{\partial x} = \frac{1}{x^2+y^2+1} \cdot (2x) = \frac{2x}{x^2+y^2+1}.$$

Similarly, the partial derivative with respect to y is

$$\frac{\partial f(x,y)}{\partial y} = \frac{2y}{x^2+y^2+1}.$$

(b) To find $\frac{\partial f(x,y)}{\partial x}$, we treat y as a constant, and use the chain rule:

1. Take the derivative of the outer function, $\sqrt{u}$, where $u \equiv xy + y^3$, which is $\frac{1}{2\sqrt{u}}$.
2. Multiply by the derivative of the inner function with respect to x, which is $\frac{\partial(xy+y^3)}{\partial x} = y$.

Putting it together, the partial derivative with respect to x is

$$\frac{\partial f(x,y)}{\partial x} = \frac{1}{2\sqrt{xy+y^3}} \cdot (y) = \frac{y}{2\sqrt{xy+y^3}}.$$

Similarly, the partial derivative with respect to y is

$$\frac{\partial f(x,y)}{\partial y} = \frac{x+3y^2}{2\sqrt{xy+y^3}}.$$